Ashish Kharche

AI Product Engineer | LLM features in mobile and web apps · TypeScript · AWS

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Summary

Software engineer who builds LLM features into mobile and web products. Built the realtime voice coach in Workout Director, a mobile workout app published on the App Store by Shelgaon Ltd. Works day to day with Grok, Grok Bot, Claude Code and Cursor through MCP. Four years at Paythru on payment apps and a TypeScript payments platform on AWS; QA on the Grok iOS and web apps at SpaceXSI.

Skills

LLM applications: realtime voice over WebRTC, client-applied tool calls with Undo and Confirm, server-minted short-lived client secrets, on-device fallback
Agents and tooling: MCP, Grok Bot, Claude Code, Cursor; agents running CLIs (Wrangler, Firebase, AWS CLI)
On-device vision: MediaPipe pose tracking, Apple Vision fallback, rep-counting state machine
Languages and cloud: TypeScript, Node.js, React, Kotlin, Jetpack Compose; AWS Lambda, API Gateway, DynamoDB, Secrets Manager, CDK, ECS Fargate, RDS PostgreSQL, SQS
Testing and release: Grok app regression testing, replay tests on pose recordings, Maestro, Vitest, JUnit, PostHog feature flags, TestFlight

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Experience

Paythru Limited, London · Mar 2024 to Oct 2026
Senior Mobile Engineer

Paythru Limited, London · Oct 2022 to Feb 2024
Mobile Engineer
EV charging and payments · iOS and Android apps, then AWS platform work · product and engineering team of 10

- In 2026, worked on Paythru's new TypeScript payments platform on AWS, designed for 10,000 terminals (CDK, ECS Fargate, RDS PostgreSQL, Lambda, SQS).
- Worked with AI-assisted development under strict review: every change checked against the AWS and Adyen docs, tested for failure paths and duplicate or out-of-order webhooks, and personally approved before deploy.

- Ensured retried or duplicate Adyen webhooks were never processed twice: idempotency keys and HMAC-checked, de-duplicated webhooks, covered by failure-path tests.
- Built EV charging features for six white-label apps with about 100,000 sessions in Kotlin and Jetpack Compose, and integrated Adyen in-app and terminal payments.

Maestro (maestro.dev), mobile UI testing framework · Mar 2023 to Sep 2024
Developer (contract, alongside Paythru)

- Wrote about 10 mobile test suites covering four core features for Maestro Cloud and 17 posts for the Maestro blog (maestro.dev/blog).

SpaceX Super Intelligence (SpaceXSI), formerly xAI · Part-time · Jan 2026 to Oct 2026
Quality Assurance Engineer

- Carried out regression and release testing on more than 500 builds of the Grok iOS and web apps across chat, Imagine, Build and MCP; gave release sign-off and tracked bugs in Linear.

Independent projects

Workout Director (published on the App Store by Shelgaon Ltd) · Live since Oct 2026

- A mobile workout app that counts reps on the device from the camera and coaches by voice. apps.apple.com/gb/app/id6813645709
- Built the voice coach (Live Coach): a WebRTC call to a realtime voice model that never receives video and learns the workout state from short text notes on the data channel.
- The model changes nothing itself: a request such as "rest for 90 seconds" returns a tool call the app applies, with Undo; larger edits wait for a Confirm.
- A Lambda function checks the feature flag and the user's limits, reads the provider key from Secrets Manager and mints a short-lived client secret per call; the app falls back to on-device speech recognition and a text reply.
- Counts reps with MediaPipe pose tracking at up to 30 frames a second; video is never recorded or uploaded, and each counter change is replayed against pose recordings before push.
- Ran the work as a set of AI agents in Grok Bot, each owning one area, with every release or paid change personally approved.

Kalo (published on the App Store by Shelgaon Ltd) · On the App Store

- A mobile nutrition app that tracks calories and macros, with photo, voice and barcode logging through a server-side proxy. apps.apple.com/gb/app/id6752915724

Education

MSc Software Engineering, Kingston University, London · Sep 2021 to Sep 2022
Bachelor of Engineering, Computer Science, Government College of Engineering Aurangabad · 2016 to 2020

Certifications

AWS Certified Solutions Architect – Associate (expired Feb 2025)
AWS Certified Cloud Practitioner (expired Feb 2025)